Daanish Islam

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OBJECTIVE

Graduate Mechanical Engineering student specializing in automation, controls and robotics, seeking a mechanical engineering role to apply design, system integration and fabrication experience across product development and advanced manufacturing

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY | Atlanta, GA
Master of Science in Mechanical Engineering *Aug 2026 – Dec 2027 (Expected)*
Coursework: Linear Control Systems, Advanced Control System Design & Implementation, Mechatronics, Engineering Design

Bachelor of Science in Mechanical Engineering | GPA: 3.66 (Highest Honour) *Aug 2022 – May 2026*

EXPERIENCE

Redwood Materials Inc. | Reno, NV | *Mechanical Engineering Intern* *May 2025 – Aug 2025*

- Designed mechanical assemblies, including a hardware mount, industrial dust-collection system, and high-load support cart, in SolidWorks; validated structural integrity with ANSYS FEA, applied GD&T, sheet-metal and weldment design, and supported hands-on assembly to meet high-load requirements for continuous manufacturing
- Programmed a FANUC robot via teach pendant for HVOF thermal spray coating on flanges and plates; used FANUC ROBOGUIDE to simulate and validate spray paths and parameters, improving coating cycle time by **~12%**
- Conducted metallurgy-based testing to verify coating quality, including microhardness, surface roughness, tensile, and adhesion tests, ensuring coating thickness of 200–400 microns and proper material hardness; improved testing efficiency by reducing process time from **9 to 7 hours**
- Created 2D part and assembly drawings, technical documents, Standard Work Instructions (SWIs) and Standard Operating Procedures (SOPs), which streamlined workflow and improved safety compliance for manufacturing operations

Bangladesh Steel Re-Rolling Mills Limited | Bangladesh | *Manufacturing Engineering Intern* *May 2024 – July 2024*

- Conducted troubleshooting and root cause analysis on recurring failures in hydraulic, pneumatic, and cooling systems, identifying valve wear and pressure inconsistencies, contributing to a **21%** reduction in unplanned downtime
- Designed and fabricated replacement valves, jigs, and lifting fixtures in SolidWorks; conducted CFD analysis to optimize flow performance, improve maintenance accessibility, and reduce equipment servicing time
- Analyzed procurement and material flow delays by tracking order processing and delivery timelines under ISO 9001 procedures, identifying bottlenecks in vendor coordination that reduced equipment replacement lead time by **3-5 days**

The Water – Energy Research Lab | Atlanta, GA | *Research Assistant, Georgia Tech* *Jan 2025 – April 2025*

- Developed and 3D printed CAD models in SolidWorks for an LCST desalination loop; integrated humidity/pressure sensors, control valves, motor and water pump, producing a functional prototype that passed initial performance validation
- Built and tested two mass exchanger configurations, using Sensirion ControlCenter to analyze desalination, mass exchange and leakage performance, achieving a relative humidity increase from **25.7% to 96.7%**

Woodruff School of Mechanical Engineering | Atlanta, GA | *Graduate Teaching Assistant* *Aug 2026 – Present*

- Led ME 3057 laboratory sections, guiding students through experiments, equipment operation and safety procedures
- Graded technical reports and provided structured feedback, contributing to gradual improvement in student course grades

Tutoring and Academic Support | Atlanta, GA | *Tutor, Georgia Tech* *Aug 2023 – May 2026*

- Tutored Math, Physics, MATLAB for **300 students**, 8 hours/week, leading to improved average course grades by **~10%**

PROJECTS

Defense Innovation Unit – US Army | Wharton, NJ | *Capstone Design, Georgia Tech* *Jan 2026 – May 2026*

- Designed a modular shock tube system in SolidWorks with driver-driven sections, diaphragm-flange assembly, and high-pressure gas booster and cascade intake system to generate repeatable **5–2000 psi** shockwaves for testing materials designed to improve soldier protection in combat
- Validated design via FEA and hand calculations using Lamé's equation with fatigue analysis for pressure-vessel and bolted-flange loading, confirming structural integrity and delivering a fully engineered CAD design for fabrication and testing

Better Place Drones | Atlanta, GA | *Research Assistant, Georgia Tech VIP Program* *Aug 2024 – Dec 2024*

- Designed and fabricated a deployable tether spool system for an X8 UAV using SolidWorks, selecting structural members and fasteners based on load calculations to support continuous power transmission while preventing cable entanglement
- Integrated mechanical frame design with power electronics and motor controls by coordinating mounting layouts, wire routing, and structural reinforcement, enabling stable **30+ minute** tethered flight within motor temperature and load limits

SKILLS

CAD & Engineering:	SolidWorks, Siemens NX, AutoCAD, ANSYS, 3D CAD, GD&T, BOM, FEA, CFD, DFM, FMEA
Automation & Controls:	FANUC Robotics, ROBOGUIDE, MATLAB, LabVIEW, DAQ, Arduino
Programming & Data:	Python, C++, MATLAB, Minitab, SAS, SPC, SQL, GitHub, Jira, Microsoft Office
Manufacturing:	CNC Machining, 3D Printing, Laser Cutting, Sheet Metal, Welding, Casting, Molding, R&D